

Part 2: Job Postings Analysis - Role Categorization and Requirements Extraction

This notebook uses **LangChain** with a **local Ollama model** (`llama3.2` , a small language model) to analyze job postings and produce, for each posting:

1. **Job Category Classification** — a broad domain (Technology/IT, Finance, Marketing, Healthcare, Education, etc.)
2. **Key Requirements Extraction** — required skills/technologies, education level, and experience

As in Part 1, we use Ollama rather than a hosted API to avoid rate limits and to make the full-dataset bonus feasible.

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```
In [1]: import json
import threading
from concurrent.futures import ThreadPoolExecutor
from pathlib import Path

import pandas as pd
from typing import List, Literal
from pydantic import BaseModel, Field
from tqdm.auto import tqdm

from langchain_ollama import ChatOllama
from langchain_core.prompts import ChatPromptTemplate
from langchain_core.output_parsers import JsonOutputParser

MODEL_NAME = "llama3.2" # local SLM served via Ollama
OLLAMA_URL = "http://localhost:11434" # explicit address: an OLLAMA_HOST of 0.0.0.0 is not a valid client address
# num_predict caps each reply so a looping generation cannot stall a long run
llm = ChatOllama(model=MODEL_NAME, temperature=0, base_url=OLLAMA_URL, num_predict=512)

print(f"Using Ollama model: {MODEL_NAME}")
```

Using Ollama model: llama3.2

Step 1: Load the Dataset

Job postings dataset with `Job Title` and `Job Description` columns, scraped from online job boards.

```
In [2]: raw_df = pd.read_csv("data/job_title_des.csv", index_col=0)
print(raw_df.shape)
raw_df.head(3)
```

(2277, 2)

```
Out[2]:
```

	Job Title	Job Description
0	Flutter Developer	We are looking for hire experts flutter develo...
1	Django Developer	PYTHON/DJANGO (Developer/Lead) - Job Code(PDJ ...
2	Machine Learning	Data Scientist (Contractor)\r\n\r\nBangalore, ...

```
In [3]: df = raw_df.rename(columns={"Job Title": "Job_Title", "Job Description": "Job_Description"}).copy()
df = df.head(25).reset_index(drop=True)
print(df.shape)
df.head()
```

(25, 2)

```
Out[3]:
```

	Job_Title	Job_Description
0	Flutter Developer	We are looking for hire experts flutter develo...
1	Django Developer	PYTHON/DJANGO (Developer/Lead) - Job Code(PDJ ...
2	Machine Learning	Data Scientist (Contractor)\r\n\r\nBangalore, ...
3	iOS Developer	JOB DESCRIPTION:\r\n\r\nStrong framework outsi...
4	Full Stack Developer	job responsibility full stack engineer – react...

Step 2: Job Category Classification

A prompt (with a few labeled examples) that maps a job title + description to a broad domain category, using `with_structured_output` for a clean single label. "Other" is used as a fallback when the role doesn't clearly fit.

```
In [4]: CATEGORIES = [
        "Technology/IT", "Finance", "Marketing", "Healthcare", "Education",
```

```

    "Sales", "Human Resources", "Operations", "Design", "Customer Service", "Other",
]

class JobCategory(BaseModel):
    category: Literal[tuple(CATEGORIES)] = Field(description="The single best-fitting broad domain for this job")

classification_prompt = ChatPromptTemplate.from_messages([
    ("system",
     "Given a job title and description, categorize the job into one of the following domains: "
     f"{'', '.join(CATEGORIES)}. If unsure, use 'Other'. Respond with only the single best-fitting category.\n\n"
     "Examples:\n"
     "Job: Software Engineer. Description: Build and maintain backend services using Java and Spring Boot.\n"
     "Domain category: Technology/IT\n\n"
     "Job: Staff Accountant. Description: Prepare financial statements and reconcile accounts monthly.\n"
     "Domain category: Finance\n\n"
     "Job: Registered Nurse. Description: Provide patient care and administer medication in a hospital ward.\n"
     "Domain category: Healthcare"),
    ("human", "Job: {title}. Description: {description}\n\nDomain category:")
])

classification_chain = classification_prompt | llm.with_structured_output(JobCategory)

# --- Test on a sample datapoint ---
sample = df.iloc[0]
sample_result = classification_chain.invoke({"title": sample["Job_Title"], "description": sample["Job_Description"][:3000]})
print("Job Title:", sample["Job_Title"])
print("Predicted category:", sample_result.category)

```

Job Title: Flutter Developer

Predicted category: Technology/IT

Step 3: Key Requirements Extraction

A single composite prompt extracts **Skills**, **Education**, and **Experience** in one call, using **JSON-mode prompting** (Ollama's `format="json"`) with a Pydantic schema validated via `JsonOutputParser`. In testing, LangChain's tool-calling `with_structured_output` frequently returned empty/blank fields on longer descriptions with this small model — JSON-mode prompting proved much more reliable. Missing information is explicitly represented as `"Not specified"` rather than left to guesswork.

```

In [5]: class JobRequirements(BaseModel):
        skills: List[str] = Field(default_factory=list, description="Key skills, technologies, or tools mentioned; empty list if

```

```

none")
    education: str = Field(description="Minimum/preferred education level, e.g. Bachelor's degree in CS; use 'Not specified' if
not mentioned")
    experience: str = Field(description="Years of experience or experience level required, e.g. '3+ years'; use 'Not specified'
if not mentioned")

requirements_parser = JsonOutputParser(pydantic_object=JobRequirements)

extraction_prompt = ChatPromptTemplate.from_messages([
    ("system",
     "Extract the required skills/technologies, minimum education level, and required years of experience "
     "from the job description below. If a piece of information is not mentioned, use 'Not specified' "
     "(for education/experience) or an empty list (for skills). Do not invent information.\n{format_instructions}"),
    ("human", "Job Title: {title}\nJob Description:\n{description}")
]).partial(format_instructions=requirements_parser.get_format_instructions())

# JSON-mode (format="json") makes the Ollama model emit valid JSON directly, which this small
# model handles more reliably than LangChain's tool-calling with_structured_output.
requirements_llm = ChatOllama(model=MODEL_NAME, temperature=0, format="json", base_url=OLLAMA_URL, num_predict=512)
extraction_chain = extraction_prompt | requirements_llm | requirements_parser

# --- Test on a sample datapoint ---
sample_reqs = extraction_chain.invoke({"title": sample["Job_Title"], "description": sample["Job_Description"][:8000]})
print("Job Title:", sample["Job_Title"])
print("Requirements:", sample_reqs)

```

Job Title: Flutter Developer

Requirements: {'skills': [], 'education': 'Not specified', 'experience': 'Not specified'}

Step 4 & 5: Apply to All Postings and Update the DataFrame

```

In [6]: def analyze_posting(title: str, description: str) -> dict:
        desc = str(description)[:8000] # requirements are often listed near the end of long postings
        try:
            category = classification_chain.invoke({"title": title, "description": desc[:3000]}).category
        except Exception:
            category = "Other"
        try:
            reqs = extraction_chain.invoke({"title": title, "description": desc})
            skills = reqs.get("skills") or []
            if isinstance(skills, str): # e.g. "Python, SQL" instead of a list

```

```

        skills = [s.strip() for s in skills.split(",") if s.strip()]
        education = reqs.get("education") or "Not specified"
        experience = reqs.get("experience") or "Not specified"
        if isinstance(education, list):
            education = ", ".join(map(str, education))
        if isinstance(experience, list):
            experience = ", ".join(map(str, experience))
    except Exception:
        skills, education, experience = [], "Not specified", "Not specified"
    return {
        "Predicted_Category": category,
        "Required_Skills": skills,
        "Education_Required": education,
        "Experience_Required": experience,
    }

```

```

def run_pipeline(data: pd.DataFrame, cache_path: str | None = None, workers: int = 1) -> pd.DataFrame:
    """Analyze every posting. With `cache_path`, each finished row is saved to a JSON-lines file so a
    long run (the bonus) can resume where it stopped. Rows with nothing extracted are retried once.
    `workers` > 1 sends several postings to Ollama at once (it serves requests in parallel)."""
    done = {}
    if cache_path and Path(cache_path).exists():
        for line in Path(cache_path).read_text(encoding="utf-8").splitlines():
            record = json.loads(line)
            done[record["row"]] = record["result"]
    todo = [(int(idx), row["Job_Title"], row["Job_Description"]) for idx, row in data.iterrows()
            if int(idx) not in done]
    cache = open(cache_path, "a", encoding="utf-8") if cache_path else None
    lock = threading.Lock()

    def work(item):
        key, title, description = item
        result = analyze_posting(title, description)
        if not result["Required_Skills"] and result["Education_Required"] == result["Experience_Required"] == "Not specified":
            result = analyze_posting(title, description) # retry once
        with lock:
            done[key] = result
            if cache:
                cache.write(json.dumps({"row": key, "result": result}) + "\n")
                cache.flush()

```

```

with ThreadPoolExecutor(max_workers=workers) as pool:
    list(tqdm(pool.map(work, todo), total=len(todo), desc="Analyzing job postings"))
if cache:
    cache.close()
results = [done[int(k)] for k in data.index]
results_df = pd.DataFrame(results)
return pd.concat([data.reset_index(drop=True), results_df], axis=1)

final_df = run_pipeline(df)
final_df.head()

```

Out[6]:

	Job_Title	Job_Description	Predicted_Category	Required_Skills	Education_Required	Experience_Required
0	Flutter Developer	We are looking for hire experts flutter develo...	Technology/IT	[]	Not specified	Not specified
1	Django Developer	PYTHON/DJANGO (Developer/Lead) - Job Code(PDJ ...	Technology/IT	[Python, Django, API development, REST/RPC, Li...	Not specified	Not specified
2	Machine Learning	Data Scientist (Contractor)\r\n\r\nBangalore, ...	Technology/IT	[Python, Java, Machine Learning, Deep Learning...	Not specified	Not specified
3	iOS Developer	JOB DESCRIPTION:\r\n\r\nStrong framework outsi...	Technology/IT	[Objective-C, Cocoa Touch, Core Data, Core Ani...	Not specified	Not specified
4	Full Stack Developer	job responsibility full stack engineer – react...	Technology/IT	[React, J Native, JavaScript, HTML, CSS, Redux...	Not specified	Not specified

In [7]:

```

final_df.to_csv("outputs/part2_jobs_first25_results.csv", index=False)
final_df.to_json("outputs/part2_jobs_first25_results.json", orient="records", indent=2)
print("Saved to outputs/part2_jobs_first25_results.csv and .json")
final_df

```

Saved to outputs/part2_jobs_first25_results.csv and .json

Out[7]:

	Job_Title	Job_Description	Predicted_Category	Required_Skills	Education_Required	Experience_Required
0	Flutter Developer	We are looking for hire experts flutter develo...	Technology/IT	[]	Not specified	Not specified
1	Django Developer	PYTHON/DJANGO (Developer/Lead) - Job Code(PDJ ...	Technology/IT	[Python, Django, API development, REST/RPC, Li...	Not specified	Not specified
2	Machine Learning	Data Scientist (Contractor)\r\n\r\nBangalore, ...	Technology/IT	[Python, Java, Machine Learning, Deep Learning...	Not specified	Not specified
3	iOS Developer	JOB DESCRIPTION:\r\n\r\nStrong framework outsi...	Technology/IT	[Objective-C, Cocoa Touch, Core Data, Core Ani...	Not specified	Not specified
4	Full Stack Developer	job responsibility full stack engineer – react...	Technology/IT	[React, J Native, JavaScript, HTML, CSS, Redux...	Not specified	Not specified
5	Java Developer	Software Developer - Integration*\r\nImmediate...	Technology/IT	[Proven technical expertise in the design, dev...	Bachelor's Degree in Computer Science, Informa...	2 years
6	Full Stack Developer	senior full stack developer \r\n1800026h cwt lo...	Technology/IT	[NodeJS, Java, MongoDB, Elasticsearch, Redis, ...	B.Sc	2 years
7	JavaScript Developer	Job Description:\r\n\r\nReactJS + NodeJS, Azur...	Technology/IT	[ReactJS, NodeJS, Azure Functions, GraphQL]	Not specified	3 - 8 years
8	DevOps Engineer	Main Responsibilities and Deliverables:\r\nMan...	Technology/IT	[Bash, Ruby, Python, Java, Puppet, Chef, Cloud...	Not specified	Not specified
9	Software Engineer	Overview\r\n\r\n\r\nBased in Silicon Valley, T...	Technology/IT	[REST API, C/C++, Python, Go, Git, Gerrit, Jen...	BS or MS; computer engineering, computer scien...	Minimum 7 years of software development experi...
10	Database Administrator	richardson texas united state bausp4 boeing comp...	Technology/IT	[SQL, Oracle, Database design, Administration,...	Bachelor's degree	Not specified
11	Machine Learning	Remote, Any where\r\n2 - 4 years\r\n₹8L - ₹12L...	Technology/IT	[TensorFlow, OpenCV, Python, Artificial Neural...	Not specified	Not specified
12	Machine Learning	Excited by Big Data, Machine Learning and Pred...	Technology/IT	[Big Data, Machine Learning, Data Mining, Java...	Masters/Ph.D.	3+ years
13	Software Engineer	As a member of our Software Engineering Group ...	Technology/IT	[HTML5, CSS 3.0, Bootstrap, Angular 8.x+, Java...	Not specified	5 - 8 years
14	Software Engineer	JOB DESCRIPTION\r\n\r\nJob Title: Software Eng...	Technology/IT	[Object-oriented design, Java, .NET developmen...	Bachelor's degree or higher education level, o...	6+ years (software development), 4 years minim...
15	Java Developer	Overview:\r\n\r\nWestern National is seeking a...	Technology/IT	[modern Javascript (ES6+), Node, Vue, React, A...	B.S. Degree in Computer Science	5+ years
16	Wordpress Developer	Experience: 2-5 years\r\n\r\nJob Location:- Au...	Technology/IT	[WordPress, Plugin-in development, PHP, HTML/H...	Not specified	2-5 years

	Job_Title	Job_Description	Predicted_Category	Required_Skills	Education_Required	Experience_Required
17	iOS Developer	Job Code : ID190704\r\n\r\nJob Description:\r\n...	Technology/IT	[iOS Development, PhoneGap, Ionic, Titanium, D...	BE (Computers/Information Technology), MCA, BC...	Not specified
18	Database Administrator	job description summary highly skilled technic...	Technology/IT	[Oracle, Oracle 10g, Oracle 11g, Oracle 12c, P...	Bachelor's degree in Computer Science or Infor...	10+ years
19	DevOps Engineer	Job Summary\r\n\r\nOur mission at Style Theory...	Technology/IT	[Linux Environment, Ansible, Chef, Puppet, Ter...	Not specified	Not specified
20	Database Administrator	country place mortgage founded 1995 subsidiary...	Finance	[Oracle, Linux, SQL, WebLogic, Database Admini...	Bachelor's degree	2+ years
21	Wordpress Developer	Position: WordPress Developer\r\n\r\nLocation: Pra...	Technology/IT	[CSS3, JavaScript, HTML5, jQuery, Git, Mercuri...	Bachelor's	4 to 8 Years
22	JavaScript Developer	MonetizeMore builds industry leading ad techno...	Technology/IT	[JavaScript, Python, Flask, PostgreSQL, Linux,...	Not specified	Not specified
23	Database Administrator	qualification essential qualification related ...	Technology/IT	[Microsoft Windows, Microsoft Office, SQL Serv...	Not specified	Not specified
24	PHP Developer	Job Summary\r\n\r\n**\r\n\r\nWe are looking for Freshe...	Technology/IT	[PHP, WordPress, HTML, CSS]	Not specified	0 - 3 Year

Verify the Outputs

Spot-check a few postings against their description text, then count how often each field was extracted across all 25 postings.

```
In [8]: for _, row in final_df.sample(3, random_state=1).iterrows():
        print(f"--- {row['Job_Title']} | category: {row['Predicted_Category']}")
        print("Description excerpt:", " ".join(str(row["Job_Description"]).split()[:350], "...")
        print("Required skills      :", row["Required_Skills"])
        print("Education required  :", row["Education_Required"])
        print("Experience required:", row["Experience_Required"], "\n")

total = len(final_df)
print(f"Skills extracted      : {(final_df['Required_Skills'].apply(len) > 0).sum()}/{total} postings")
print(f"Education mentioned   : {(final_df['Education_Required'] != 'Not specified').sum()}/{total} postings")
print(f"Experience mentioned  : {(final_df['Experience_Required'] != 'Not specified').sum()}/{total} postings")
```


--- Software Engineer | category: Technology/IT

Description excerpt: JOB DESCRIPTION Job Title: Software Engineer I SUMMARY The Software Engineer I is responsible to design, code, and/or configure solutions for moderate complexity Agile stories, as well as writing automated unit and integration-level tests. ESSENTIAL JOB FUNCTIONS/RESPONSIBILITIES Designs, codes, and/or configures solutions for moderate complexity A ...

Required skills : ['Object-oriented design', 'Java', '.NET development', 'Relational OLTP queries', 'Relational database design', 'XML/XSLT document design', 'JavaScript development', 'HTML5', 'CSS']

Education required : Bachelor's degree or higher education level, or its foreign equivalent, in Computer Science, Computer Information Sciences, and/or related field

Experience required: 6+ years (software development), 4 years minimum, Product Experience: 2 years minimum, 4+ years preferred, Domain Experience: 2 years minimum, 4+ years preferred

--- Software Engineer | category: Technology/IT

Description excerpt: As a member of our Software Engineering Group you will dive head-first into creating innovative solutions that advance businesses and careers. You'll join an inspiring and curious team of technologists dedicated to improving the design, analytics, development, coding, testing and application programming that goes into creating high quality software ...

Required skills : ['HTML5', 'CSS 3.0', 'Bootstrap', 'Angular 8.x+', 'JavaScript', 'jQuery', 'Java', 'Unit testing with Jasmine or protractor', 'Testing tools - Jasmine/ Karma/ protractor', 'Version Control - GIT', 'Micro-services pattern', 'Design principles and design patterns', 'Agile experience', 'Selenium', 'Jenkins', 'Ci/CD']

Education required : Not specified

Experience required: 5 - 8 years

--- iOS Developer | category: Technology/IT

Description excerpt: Job Code : ID190704 Job Description: Experience: 1 to 2 years in iOS Development Native iOS development Hybrid app development with PhoneGap, Ionic, Titanium will be an advantage Database knowledge is must Understanding about Requirement Gathering - Client interactions Good communication skills Qualification: BE(Computers/ Information Technology), ...

Required skills : ['iOS Development', 'PhoneGap', 'Ionic', 'Titanium', 'Database']

Education required : BE (Computers/Information Technology), MCA, BCA, BSc (Computer Science)

Experience required: Not specified

Skills extracted : 24/25 postings

Education mentioned : 11/25 postings

Experience mentioned : 13/25 postings

Bonus: Full Dataset (all job postings)

`RUN_BONUS = True` below processes every posting in the dataset instead of just the first 25. This runs locally via Ollama with no external rate limits, but will take a while (roughly 1-2 hours depending on hardware, since each posting needs 2 LLM calls: classification and requirements extraction).

Finished rows are saved to `outputs/part2_jobs_ALL_cache.jsonl` as the run goes, so an interrupted run continues where it stopped. Set `RUN_BONUS = False` to skip this cell.

```
In [9]: RUN_BONUS = True

if RUN_BONUS:
    bonus_df = raw_df.rename(columns={"Job Title": "Job_Title", "Job Description": "Job_Description"}).copy()

    bonus_results_df = run_pipeline(bonus_df, cache_path="outputs/part2_jobs_ALL_cache.jsonl", workers=4)
    bonus_results_df.to_csv("outputs/part2_jobs_ALL_results.csv", index=False)
    bonus_results_df.to_json("outputs/part2_jobs_ALL_results.json", orient="records", indent=2)
    no_skills = (bonus_results_df["Required_Skills"].apply(len) == 0).sum()
    print(f"Rows analysed: {len(bonus_results_df)} | postings with no skills extracted: {no_skills}")
    print("Saved to outputs/part2_jobs_ALL_results.csv and .json")
else:
    print("RUN_BONUS is False - skipping full-dataset run. Set RUN_BONUS = True to process every posting.")
```

Rows analysed: 2277 | postings with no skills extracted: 51

Saved to outputs/part2_jobs_ALL_results.csv and .json